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MINOR PROJECT REVIEW

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Low-Profile Dual Polarized Stepped Slot Antenna Array for Base Stations



TOPICS :

- Abstract
- Introduction
- Design Considerations
- Antenna Array Construction
- Optimization
- Installation
- Conclusion

Abstract :

- The antenna is composed of two pairs of **SI slots** excited by two **orthogonal stepped microstrip feedlines**.
- Measured results demonstrate the bandwidths (**return loss dB**) of the antenna array are **46.9% (1.55-2.5 GHz)** and **38.7% (1.69–2.5 GHz)** for Port 1 and Port 2, respectively.

Introduction:

- A dual polarized stepped slot antenna array is a type of antenna used in base station applications.
- Its design allows for the transmission of two different polarizations, **making it ideal for communication systems that require dual polarization.**
- It also has a low-profile design, making it suitable for use in **small spaces.**

Design Considerations :

- Designing a low-profile dual polarized stepped slot antenna array for base stations requires careful consideration of several factors. These include the **antenna's size, shape, and polarization**.
- Additionally, the antenna must be designed to optimize **power transfer efficiency and minimize interference**.
- The design must also take into account the **environment** in which the antenna will be used and the type of signals it will be transmitting.

Antenna Array Construction :

- The antenna array is constructed from a **series of metal plates** that are connected by a **series of metal rods**.
- The plates are arranged in a **stepped pattern**, with each plate having a **slightly different size and shape** than the one before it.
- This arrangement creates a stepped slot antenna array that is **capable of transmitting** two different polarizations.
- The metal rods are used to connect the plates and provide support for the antenna array.

Optimization :

- The antenna array must be optimized to ensure that it is capable of transmitting signals with the **highest power transfer efficiency and the lowest interference.**
- This requires careful consideration of the **antenna's size, shape, and polarization.**
- Additionally, the antenna must be designed to minimize the **amount of energy that is reflected or scattered away** from the antenna array.

Installation :

- Once the antenna array is optimized, it must be installed in its **intended environment**.
- This requires careful consideration of the environment in which the antenna will be used and the type of signals it will be transmitting.
- Additionally, the antenna must be securely mounted to a **surface and shielded** from any sources of interference.

Conclusion :

- A wideband dual-polarized stepped slot antenna with a relatively low profile is proposed in this letter. The proposed antenna achieves an **overlapped band of 620 MHz** for both polarization.
- The array prototype achieves a wide band in the range **of 1.69–2.5 GHz**, which covers **GSM, PCS, UMTS, and WLAN**.
- Over the entire operating band, the isolation between the two ports keeps **higher than 35 dB**.